

Mayte Alvarez

The Honors College | Miami Dade College

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SUMMARY

Physics Undergraduate Researcher specializing in nonlinear dynamics and computational modeling. Currently investigating deterministic chaos in solar radio bursts through phase-space reconstruction and time-series analysis.

As a first-generation immigrant with a non-traditional academic path, I combine a deep foundation in Mathematics with a dedication to promoting research initiatives and mentorship for emerging Hispano and Cuban scientists. Aiming to bridge the gap between solar atmospheric dynamics and laboratory plasma physics to accelerate the development of sustainable fusion energy.

EDUCATION

Miami Dade College — The Honors College (Padron Campus) *Expected Graduation: Spring 2026*

Associate in Arts, Physics (Dual Language Program)

Cumulative GPA: **4.0**

Completed/Ongoing Coursework: Calculus I–III, Linear Algebra, Discrete Mathematics, Discrete Structures, Differential Equations, Abstract Algebra, Physics I–II with Calculus and Lab, Chemistry I–II with Lab, General Biology I with Lab

Prior Education — Cuba

Instituto Preuniversitario Vocacional de Ciencias Exactas “Vladimir Ilich Lenin” Havana, Cuba

Selective National STEM High School 2018 – 2022

- Completed advanced coursework in mathematics, physics, and analytical reasoning
- Conducted faculty-mentored research in Multiple Intelligence Theory; presented work at academic symposia
- Selected for regional academic Olympiads in Mathematics, Physics, and Spanish

Colegio Universitario (Physics Track), Universidad de La Habana Havana, Cuba

Selective National STEM Program, Senior Year of High School 2021 – Feb 2022

- Selected through a competitive dual-exam admissions process for advanced university-level coursework
- Completed coursework with final grades $\geq 85\%$, earning direct admission to the University of Havana’s School of Physics without standard national entrance examinations

RESEARCH EXPERIENCE

Solar Radio Burst Dynamics Research

Miami, Florida

Undergraduate Researcher, Mentor: Adolfo L. Méndez, PhD

May 2025 – Present

- Analyze over 24,000 solar radio burst data points to characterize spike-type fine temporal structures and investigate the underlying dynamical behavior of solar emission processes.
- Apply nonlinear time-series methods, including phase-space reconstruction, non-linear exponents estimation, and correlation dimension analysis, to identify signatures of deterministic chaos.
- Contribute to manuscript development and dissemination of findings through poster and oral presentations at national and international scientific conferences.

Sharky Robotics Sustainability Project

Miami, Florida

Research Assistant, Mentor: Jyrko Correa, PhD

October 2024 – Present

- Co-designed an autonomous educational robot integrated with interactive features to promote environmental stewardship and climate change awareness among college community.

- Lead mechanical design, circuit development, and software implementation of robotic components, collaborating with a small interdisciplinary student team under faculty mentorship.
- Perform iterative testing and optimization to improve system performance, reliability, and suitability for educational outreach.
- Present the completed prototype at national academic and research conferences; project supported through competitive grant funding.

RESEARCH PRESENTATIONS

Solar Radio Burst Dynamics Research

- American Geophysical Union (AGU) Fall Meeting — 2025 (Poster)
- MIT Undergraduate Research Technology Conference (URTC) — 2025 (Poster)
- IFoRE Conference — 2025 (Oral Presentation)
- Miami Dade College School of Science Symposium — 2025 (Poster)
- Honors College STEM Undergraduate Research Symposium — 2025 (Poster)
- National Conference on Undergraduate Research (NCUR) — 2026 (Poster)

Sharky Robotics Sustainability Project

- American Geophysical Union (AGU) Fall Meeting — 2025 (Oral Presentation)
- National Conference on Undergraduate Research (NCUR) — 2025 (Poster)

LEADERSHIP & SERVICE

Vice President of Academic Affairs, Sigma Zeta Math & Science Honor Society-Gamma Rho Chapter

Miami Dade College, Padron Campus

Miami, Florida

February 2025 – Present

- Lead academic and research programming for STEM students, including mentorship, member training, research opportunities, and interdisciplinary events.
- Organize STEM workshops and volunteer initiatives, coordinating partnerships, cross-club collaboration, and community efforts supporting youth entrepreneurship and local food distribution.

Founder and Program Lead, Rising Leaders Cohort

Sigma Zeta Math & Science Honor Society

Miami, Florida

September 2025 – Present

- Founded and led a selective first-year STEM leadership and research-development cohort, designing the program structure, selection process, and curriculum; interviewed and selected 14 students.
- Mentored student-led research and community initiatives, including research literacy workshops, a Sigma Zeta web platform and newsletter, partnerships with The Dennis Project, College-wide Math and Science Olympiads, and early-stage projects in science outreach, fundraising, and inclusive STEM engagement.

Local Lead, NASA International Space Apps Challenge (Miami)

NASA International Space Apps Challenge

Miami, Florida

October 2025

- Led and relaunched NASA's global hackathon for the Miami site, managing logistics, outreach, team coordination, and judging; grew participation to 95+ registrants and 14 project submissions.
- Mentored teams on technical execution and data use, contributing to one team's global nomination.

Honors College Mentor

The Honors College, Miami Dade College

Miami, Florida

August 2025 – Present

- Selected by the Director of the Honors College to mentor first-year Honors students through their transition to college-level academics including academic planning, campus resources, leadership development support, and guidance in research and service opportunities.

Member and Volunteer, Phi Theta Kappa Honor Society (PTK)

Miami Dade College

Miami, Florida

September 2024 – Present

- Participate in community service initiatives supporting youth entrepreneurship (Slingshot Marketplace) and food distribution programs for local families and older adults (Commissioner Raquel Regalado).

Stamps Enrichment Fund Ambassador
The Honors College, Miami Dade College

Miami, Florida
April 2025 – Present

- Represent the Stamps Enrichment Fund as a student ambassador, co-leading informational workshops with Honors College leadership and sharing personal experiences with students and donors on the impact of enrichment funding.

TEACHING & WORK EXPERIENCE

STEM Success Center (Mathematics), Miami Dade College
Student Assistant; Tutor

Miami, Florida
June 2025 – Present

- Support 10–15 students per week through one-on-one and small-group mathematics tutoring, strengthening conceptual understanding and problem-solving skills.
- **Peer-Led Team Learning Facilitator** in Discrete Mathematics and Calculus I-II classes.
- Provide academic and instructional support, including proctoring assessments, reviewing course materials, and assisting faculty with supplemental instruction.

MedCare Home Health Services
Authorizations Reviewer

Miami, Florida
September 2023 – November 2025

- Reviewed and evaluated more than 100 patient care plans daily, optimizing authorized services and visit structures to ensure accuracy, appropriateness, and compliance with clinical and organizational standards.

HONORS & AWARDS

- Miami Dade College Nominee, **Barry Goldwater Scholarship** - January 2026
- Beta Kappa Iota Chapter Nominee, **PTK All-USA Academic Team** — November 2025
- Phi Theta Kappa Coca-Cola Leaders of Promise Scholarship — August 2025
- NASA International Space Apps Challenge, Stellar Lead — October 2025
- Stamps Enrichment Fund Recipient — April 2025
- Honors College Fellowship Award — August 2024 – 2026
- Dean's List (All Semesters) — 2024–2025
- First Place, Regional Spanish Writing Olympiad — Cuba, 2019
- First Place, Regional Geography Olympiad — Cuba, 2019
- Special Mention, Mathematics and Physics Olympiads — Cuba, 2019–2020

TECHNICAL & RESEARCH SKILLS

- **Programming & Computation:** MATLAB, Swift (Certified User and Associate), Arduino, $\text{\LaTeX}$
- **Data Analysis & Methods:** Nonlinear time-series analysis, chaos theory methods, phase-space reconstruction, correlation dimension analysis
- **Engineering & Design:** CAD (Onshape), circuit design, basic prototyping and testing, system optimization

LANGUAGES

- Spanish — Native
- English — Fluent
- French — Intermediate